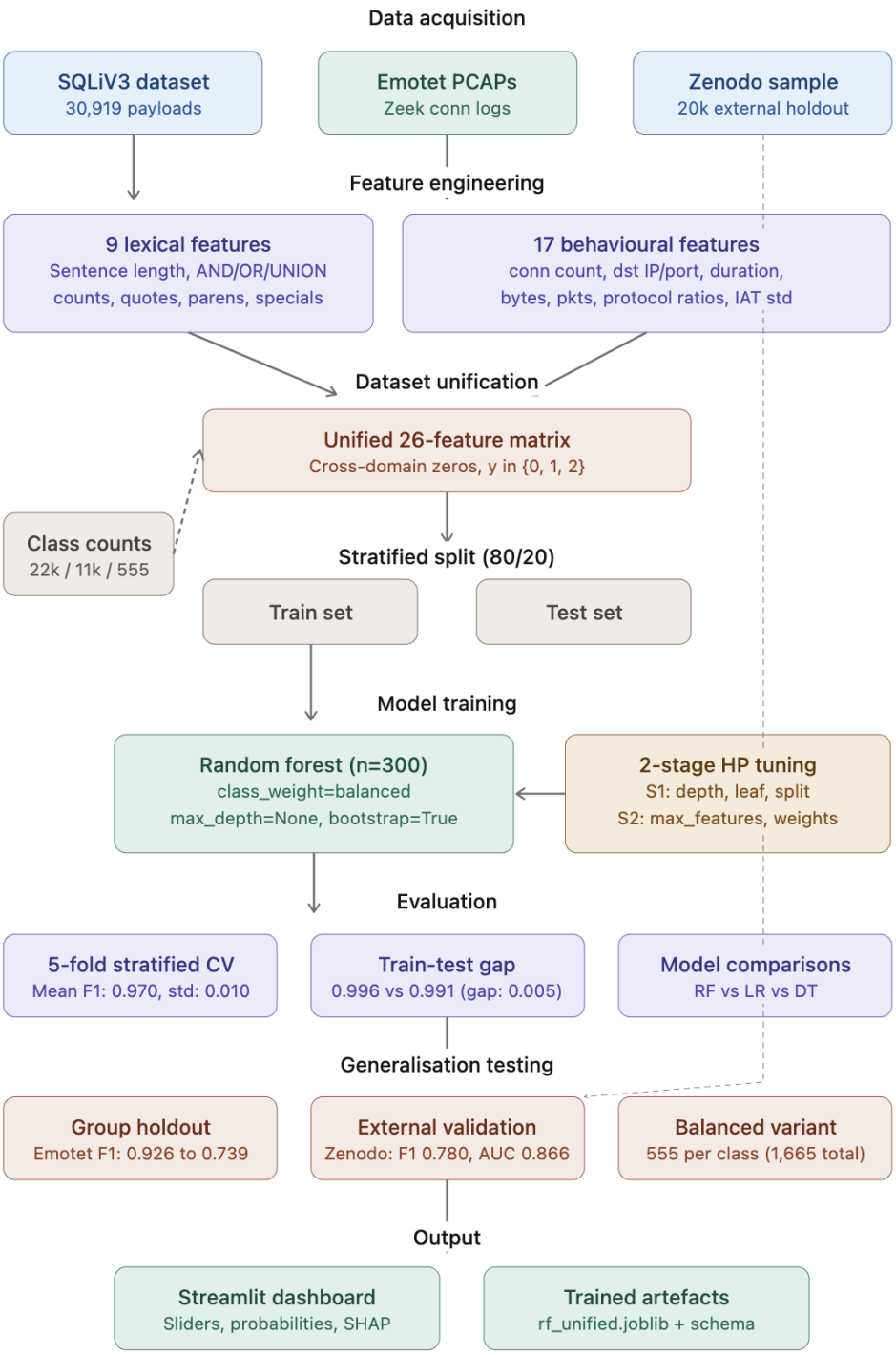
ML Model Diagram



Pipeline stages are clickable for detail.
Dashed lines indicate data flow outside the main training path.

Final: Macro-F1 0.970 | Normal 0.994 | SQLi 0.991 | Emotet 0.926

Data Acquisition: Three datasets feed the pipeline. SQLiV3 provides 30,919 raw SQL payloads for binary and multiclass training, Emotet PCAPs are processed through Zeek into connection-level behavioural logs, and a 20,000-sample Zenodo subset is held out entirely for external validation.

Feature Engineering: Two structurally distinct feature families are extracted. Nine lexical features (sentence length, operator counts, quote counts, parentheses, special characters) are derived from raw SQL query strings, and seventeen behavioural features (connection volume, destination diversity, duration statistics, protocol ratios, byte/packet summaries, rejection and reset ratios) are aggregated from Zeek connection logs over fixed time windows.

Dataset Unification: The two feature families are merged into a single 26-column matrix with three target classes: Normal (y=0), SQLi (y=1), and Emotet (y=2). Cross-domain features are zero-filled -- SQLi rows carry zeros for all Emotet columns and vice versa. The full dataset contains 34,295 rows with significant class imbalance (22,358 Normal, 11,382 SQLi, 555 Emotet), and a balanced variant with 555 samples per class was also constructed.

Stratified Split: An 80/20 stratified train-test split preserves the original class distribution in both subsets, ensuring that the minority Emotet class is proportionally represented in training and evaluation.

Model Training: The classifier is a Random Forest with 300 decision trees, class_weight="balanced" to penalise misclassification of underrepresented classes, and bootstrap aggregation. Hyperparameter tuning was conducted in two stages using 5-fold stratified cross-validation on the training set only -- Stage 1 searched structural regularisation parameters (max_depth, min_samples_leaf, min_samples_split), Stage 2 held Stage 1 winners fixed and evaluated max_features and class_weight. The tuned configuration showed marginal internal improvement but degraded external performance, so the baseline configuration was retained as the final model.

Evaluation: Internal performance was assessed through 5-fold stratified cross-validation (mean Macro-F1: 0.970, std: 0.010), train-test gap analysis (0.996 vs 0.991, gap of 0.005), and model comparisons against a Decision Tree (higher CV variance) and Logistic Regression (collapsed on Emotet with F1 of 0.501).

Generalisation Testing: Three strategies target different leakage and distribution shift risks. Group-aware holdout withheld entire PCAP captures from training to prevent Emotet data leakage, revealing a performance drop from 0.926 to 0.739 F1 on Emotet. External validation tested the SQLi component on the Zenodo dataset never seen during training, producing an F1 of 0.780 and ROC-AUC of 0.866 -- the drop reflecting distribution shift from different payload structures and obfuscation styles. The balanced dataset variant (555 per class) was used for supplementary comparison experiments.

Output: The pipeline produces serialised model artefacts (rf_unified_multiclass.joblib with a frozen feature schema) and an interactive Streamlit dashboard providing feature-level sliders, real-time per-class probability visualisations, and feature importance displays for model explainability.